

ITIS 4250
Efren Antonio
Lab #2
October 2, 2024

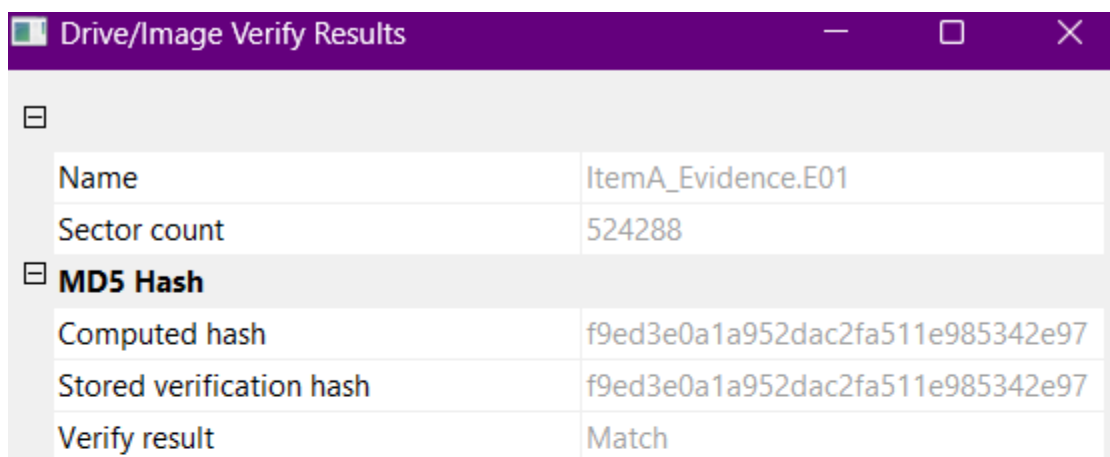
Overview

On October 2, 2024, Det. Lansbury of the UNCC Police Department provided me with a forensic image of a thumb drive for analysis. The image was taken by Officer Jimmy Johns of the UNCC Police Department, who is investigating a case involving stolen Apple devices. The thumb drive was obtained with the subject's consent, and it is suspected to contain evidence related to the theft of various Apple products including Macbooks, iMacs and Mac Minis.

Officer Johns also seized several other items, including an iPhone 6s, iPads, a Mac Mini, and an Apple laptop from the subject's possession. Officer Johns asked me to identify any photographs or other evidence stored on the thumb drive that may be relevant to the ongoing investigation of stolen Apple devices.

Exam Preparation

The examination environment used is a HP Laptop running Microsoft Windows 11 Home Version 10.0.22631. The tool used for verification of the image is Forensic Tool Kit (FTK) Imager Version 4.7.1.2. The tool used for recovering files and analyzing image metadata is Autopsy Version 4.21.0. The verified hash of the image is **f9ed3e0a1a952dac2fa511e985342e97** and is provided below:



Drive/Image Verify Results	
[-] [] [X]	
[-]	
Name	ItemA_Evidence.E01
Sector count	524288
[-] MD5 Hash	
Computed hash	f9ed3e0a1a952dac2fa511e985342e97
Stored verification hash	f9ed3e0a1a952dac2fa511e985342e97
Verify result	Match

Figure 1: FTK Imager MD5 Hash Verification

Analysis

a. Photographs of Apple devices:

A review of the thumb drive's contents revealed a folder named CarvedFiles which included deleted files that Autopsy recovered. Inside the CarvedFiles folder is a file named **f0017230.jpg** which contains a photograph of a Apple Mac Mini device with a sticker under the logo.

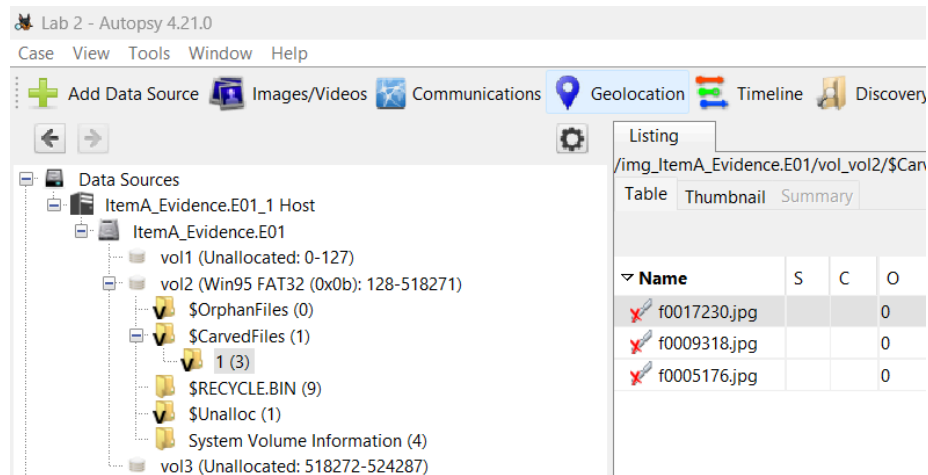


Figure 2: CarvedFiles Folder Where f0017230.jpg Is Located

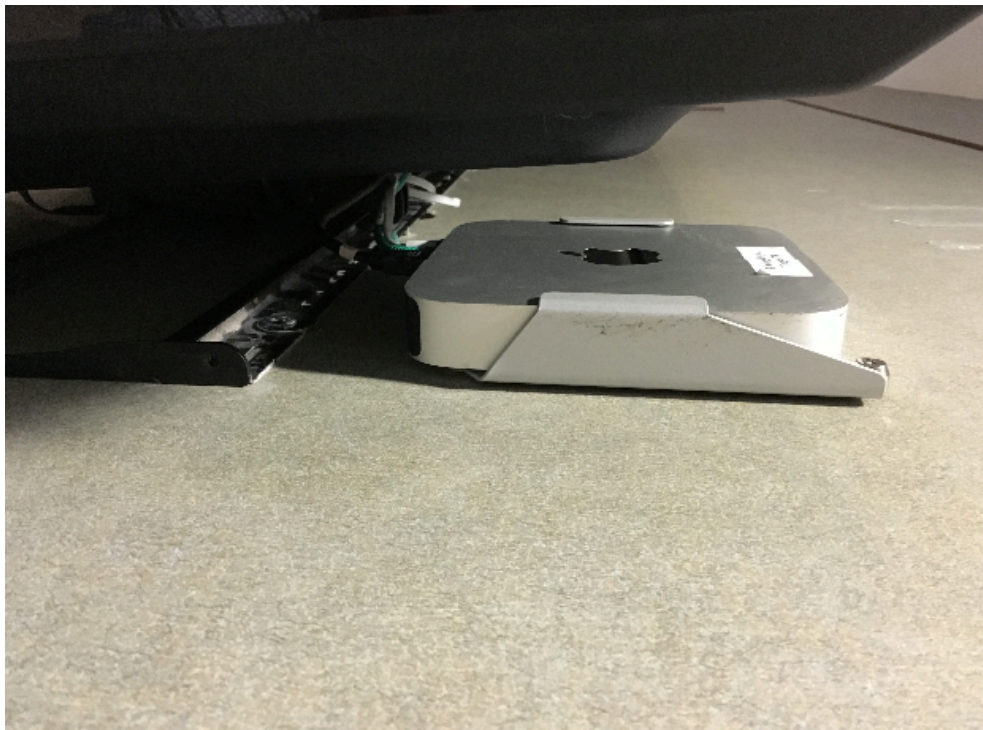


Figure 3: f0017230.jpg

b. Location information of those devices:

Using the metadata from f0017230.jpg and the geolocation feature in Autopsy, the location where the photograph was taken could be found. The coordinates of **35.307144, -80.736389** puts the device around **Woodward Hall at the University of North Carolina at Charlotte**.

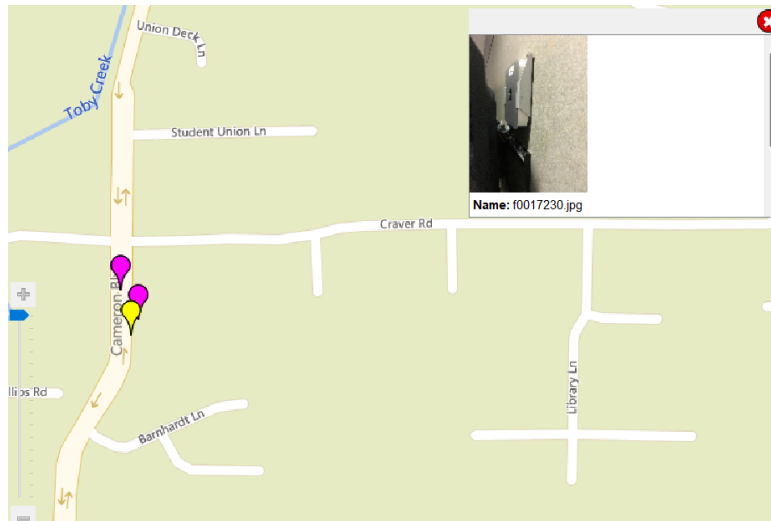


Figure 4: Autopsy Geolocation Feature Showing Location Of f0017230.jpg

Latitude:	35.30714444444444
Longitude:	-80.73638888888889

Figure 5: Coordinates In Metadata Of f0017230.jpg

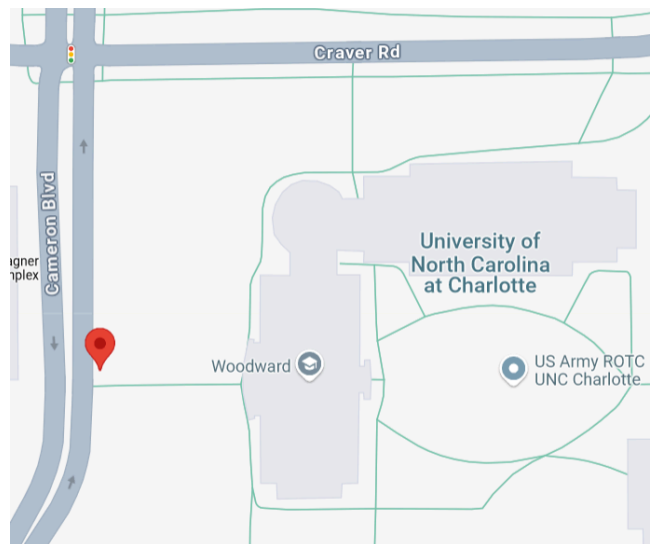


Figure 6: Coordinates Put Into Google Maps Showing Woodward Hall Location

c. Make/model of the camera device used to make the photos:

The make and model of the camera device used to make the photo is an **Apple iPhone 6s**. This was obtained from the metadata from the f0017230.jpg file.

Details	
	
Attribute	Value
	0a51cf
 Hashset	
 Camera Make	Apple
 Camera Model	Iphone 6s

Figure 7: Metadata Of f0017230.jpg Showing Camera Make/Model

d. Any content in the picture that might help identify the device:

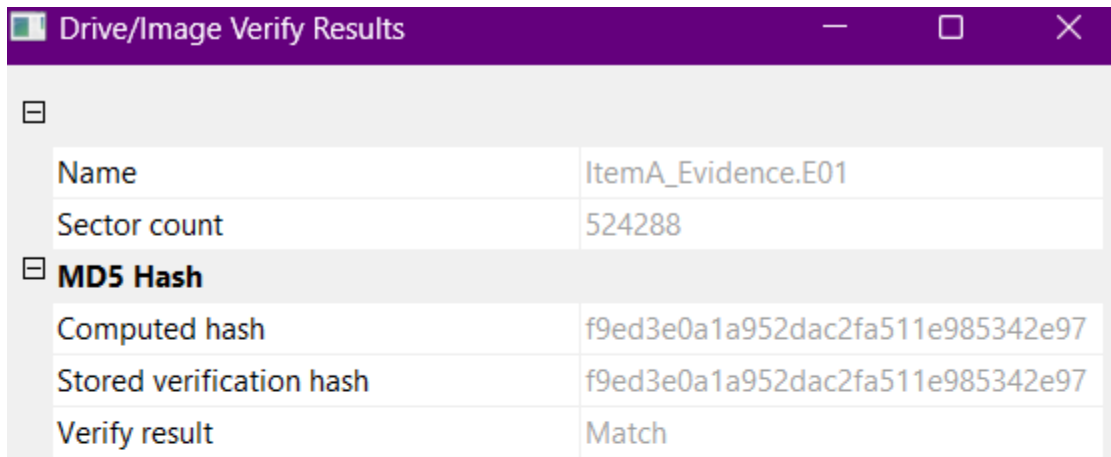
The **Apple Mac Mini** in the picture has a **distinct sticker underneath the Apple logo**. This matches the description in the report provided by Officer Johns about a Mac Mini with a sticker scraped off. This can be seen in the image below:



Figure 8: Sticker Underneath Apple Logo

Conclusion

The forensic analysis of the thumb drive provided by Officer Johns has uncovered a photograph of an Apple Mac Mini that matches the description of an item reported stolen from Woodward Hall. Using the image metadata and features in Autopsy, the location of where the image was taken could be traced back to the area around Woodward Hall at the University of North Carolina at Charlotte. The image metadata also reveals that the image was taken from an Apple iPhone 6s camera.



Drive/Image Verify Results	
[-] [] [X]	
[-]	
Name	ItemA_Evidence.E01
Sector count	524288
[-] MD5 Hash	
Computed hash	f9ed3e0a1a952dac2fa511e985342e97
Stored verification hash	f9ed3e0a1a952dac2fa511e985342e97
Verify result	Match

Figure 9: Reverified MD5 Hash Value After The Examination

Signed:

Efren Antonio